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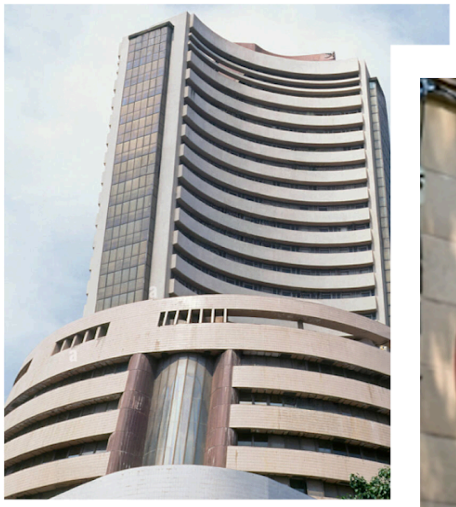
General Knowledge

Indian Economy

STUDY NOTES

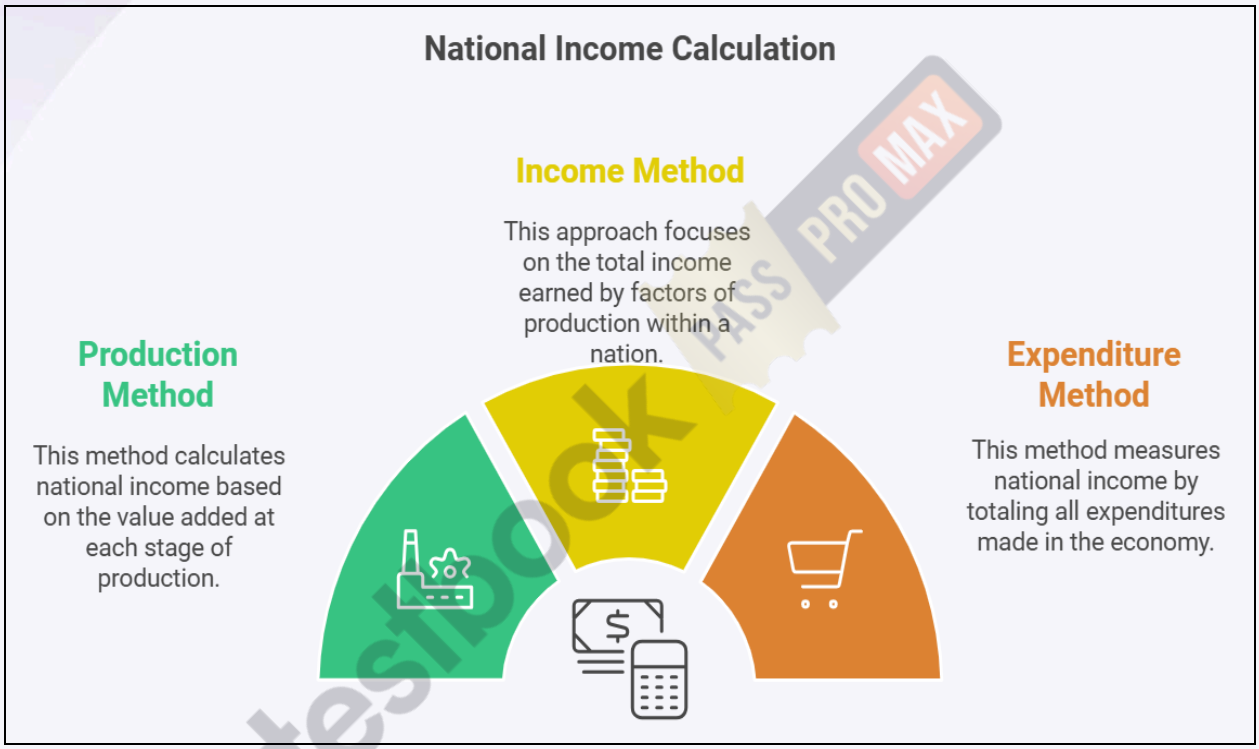
Methods of Calculating National Income



METHODS OF CALCULATING NATIONAL INCOME

National income refers to the total value of all **goods** and **services** produced in a country over a specific period, typically one year. It is an important indicator of the economic performance of a country.

Methods of Calculating National Income:



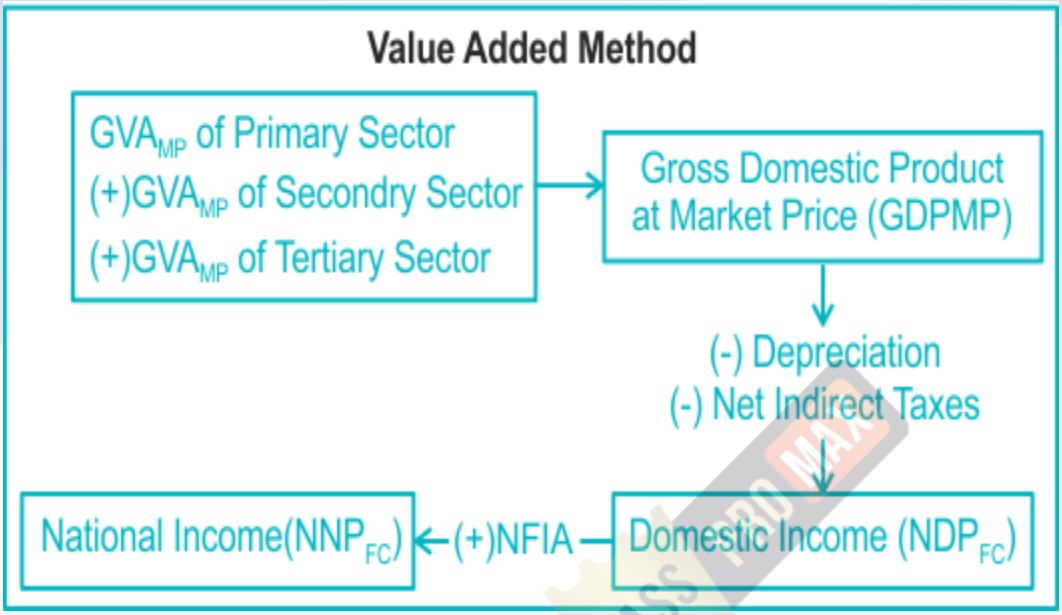
Production Method (Value Added Method):

This method calculates national income based on the value of goods and services produced by the different sectors of the economy during a given period.

Steps to Calculate National Income using the Production Method:

- Identify Sectors: Divide the economy into primary (agriculture), secondary (industry), and tertiary (services) sectors.

- Calculate Gross Value of Output (GVO): The GVO for each sector is calculated by summing up the market value of all final goods and services produced in that sector.



- Subtract Intermediate Consumption: Intermediate consumption refers to the value of goods and services used as inputs in the production process (raw materials, fuel, etc.).
    - National income is calculated by subtracting the value of intermediate goods from the gross value of output to avoid double-counting. This gives the Net Value Added.
- Sum the Net Value Added: Add the net value added from all sectors to get the Gross Domestic Product (GDP) at factor cost.

Formula: GDP (at Factor Cost) = Σ(Gross Value of Output−Intermediate Consumption)

Example:

If the agriculture sector produces ₹1,000 worth of crops, and it uses ₹300 worth of seeds and fertilizers (intermediate goods), then the net value added in agriculture is ₹700.

Key Points:

- Value Added: The difference between the value of output and intermediate consumption.
- Only final goods and services are considered to avoid double-counting.

- This method is useful for comparing the contributions of different sectors to the national income.

Income Method:

This method calculates national income by adding up all the incomes earned by individuals and businesses in the economy during the production of goods and services.

Steps to Calculate National Income using the Income Method:

- Identify Factor Incomes:
  - Factor incomes refer to the incomes earned by the factors of production (land, labour, capital, and entrepreneurship).
- Components of Factor Incomes:
  - Wages and Salaries: Income earned by labour.
  - Rent: Income earned by land.
  - Interest: Income earned by capital.
  - Profits: Income earned by entrepreneurs.
  - Mixed Income: Income earned by self-employed individuals (like small traders or farmers) where it's difficult to separate income into distinct categories.
- Calculate National Income: Sum all factor incomes (wages, rent, interest, and profits) to get the national income.

Formula: National Income = Wages + Rent + Interest+ Profits + Mixed Income

Example:

If the total wages in an economy are ₹10,000, rent is ₹5,000, interest is ₹3,000, and profits are ₹2,000, the total national income would be ₹20,000.

Key Points:

- This method focuses on income earned by individuals and businesses.
- Only factor incomes that arise from the production process are included. Transfer payments (like pensions, and scholarships) are not included as they are not earned from production.
- Useful for analyzing the distribution of income among different factors of production.

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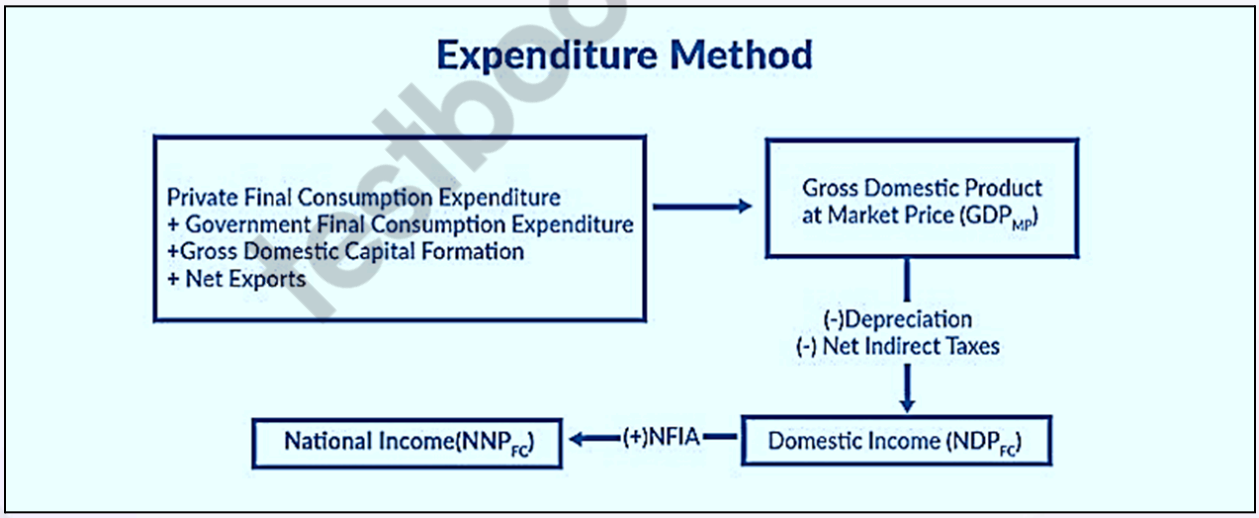
Expenditure Method:

This method calculates national income by adding up all the expenditures made in an economy on final goods and services during a specific period.

Components of Expenditure:

- Private Consumption Expenditure (C): The total amount spent by households on goods and services for personal use (food, clothing, health, etc.).
- Government Expenditure (G): The total amount spent by the government on goods and services (infrastructure, defence, education, etc.).
- Investment or Gross Domestic Capital Formation (I): Total spending by businesses on capital goods like machinery, buildings, etc. (This includes inventory investment as well).
- Net Exports (X - M): The difference between exports (X) and imports (M). Exports are added because they represent spending by foreign buyers on domestic goods, while imports are subtracted because they represent spending on foreign goods.



Formula: GDP (at Market Prices) = C+G+I+(X−M)

When the government and the external sector are present in the economy, then the savings and investment may vary but an increase in savings generally leads to an increase in the investments and the circular flow will always hold true.

Whatever income households receive, either they spend it for consumption or save it (S) or pay taxes (T).

So,

GDP = C + S + T

C + I + G + X - M = C + S + T I + G + X - M = S + T

(I - S) + (G-T) = M -X

If there is no govt. and no external sector then, G = T = M = X = 0 Hence,

I = S

Example

If a country's private consumption is ₹50,000, government expenditure is ₹10,000, investment is ₹20,000, and net exports are ₹5,000, the GDP is ₹85,000.

Key Points:

- Focuses on final expenditure to avoid double-counting.
- Includes both domestic and foreign demand for the country's goods and services.
- Shows how the national income is spent by different sectors (households, government, businesses, and foreigners).

Relationship Between Methods:

All three methods are interconnected, and they ultimately yield the same national income figure when measured correctly:

- Production Method focuses on the output side.
- The income Method focuses on the income generated from production.
- The expenditure Method focuses on the spending on that output.

The three methods highlight different aspects of economic activity but are complementary.

Important Adjustments to National Income:

- Net Factor Income from Abroad (NFIA):

- It refers to the difference between income earned by a country's residents from abroad and income earned by foreigners within the country.
- To convert GDP to GNP, we add NFIA.

GNP = GDP + NFIA

- Depreciation:
  - Depreciation refers to the reduction in the value of capital assets over time due to wear and tear. To convert Gross measures to Net measures, we subtract depreciation.
- Net National Product (NNP) = GNP – Depreciation
- Indirect Taxes and Subsidies:
  - Indirect taxes (like GST) are added to factor costs to get market prices, while subsidies are subtracted to convert market prices to factor costs.

Important Formulae:

GDP at Factor Cost = Total Value Added by all sectors.

GNP (Gross National Product) = GDP + Net Factor Income from Abroad.

NNP (Net National Product) = GNP - Depreciation.

NNP at Factor Cost (National Income) = NNP at Market Price - Indirect Taxes + Subsidies.

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